Homework 4, Math 609

October 8, 2026

PROBLEM 1: (10 points) Prove that the expression

$$\|v\|_B := \sqrt{v^* B v}$$

defines a norm on $\mathbb{C}^n$, provided that B is Hermitian and positive definite.

PROBLEM 2: (10 points) Let A be $n \times n$ matrix, not assumed to be symmetric or positive definite. Assume the existence of an A-orthonormal system $\{u^{(1)}, \ldots, u^{(n)}\}$, that is

$$(u^{(i)}, u^{(i)})_A = 1, \quad i = 1, \ldots, n, \quad \text{and} \quad (u^{(i)}, u^{(j)})_A = 0, \quad i \neq j, \quad i, j = 1, \ldots, n.$$

Prove that A is symmetric and positive definite.

PROBLEM 3: (10 points) Suppose that $Au = b$, where $A^T = A$. Let $r := Av - b$ and $e := u - v$. Prove that

$$2Q_A(v) = (e, e)_A - (u, u)_A,$$

where

$$Q_A(v) := \frac{1}{2}(v, v)_A - (b, v)_I, \quad (b, v)_I = b^T v, \quad (u, v)_A = u^T A v.$$

PROBLEM 4:

- **(10 points)** Code and test the GD method AND the CG method in the case $A = (a_{ij})$, where $a_{ij} = \frac{1}{1+i+j}$ and $b = (b_i)$, with $b_i = \frac{1}{3}\sum_{j=1}^{n} a_{ij}$, where A is 16×16 and 32×32 matrix and $x^{(0)} = 0$.

- **(10 points)** Solve the system $Ax = b$, where

$$A = \begin{pmatrix} 10 & 1 & 2 & 3 & 4 \\ 1 & 9 & -1 & 2 & -3 \\ 2 & -1 & 7 & 3 & -5 \\ 3 & 2 & 3 & 12 & -1 \\ 4 & -3 & -5 & -1 & 15 \end{pmatrix},$$

  and $b^T = (12, -27, 14, -17, 12)$, using (a) Jacobi, (b) Gauss-Seidel, (c) GD method, starting with $x^{(0)} = 0$, (d) CG method, starting with $x^{(0)} = 0$